

A

Online MLLM Editing

Edit request



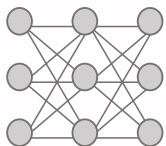
Q: Who walked on the moon first?

Target answer:
Neil Armstrong

**B**

Existing focus

MLLM editor
(existing methods)



Reliability
(edit success)



Long-horizon stability

Lack of explicit scope control

(i) Under-generalization



Paraphrased question (in-scope)

Q: Who was the first person to set foot on the Moon?

Predicted: Unknown



(ii) Over-generalization



Out-of-scope question

Q: Who was the first human in space?

Predicted: Neil Armstrong
(should be Gagarin)

**C**

Our formulation: Edit-Scoped Generalization

(1) In-scope cross-modal generalization (semantic region: "first person on the Moon")

Original edit (text)

Q: Who walked on the moon first?

→ **Armstrong** ✓

Paraphrase (text)

Q: Who was the first to walk on the Moon?

→ **Armstrong** ✓

Alt. image



→ **Armstrong** ✓

Alt. view



→ **Armstrong** ✓

Edited knowledge

(2) Out-of-scope locality preservation (outside the semantic region)

Unrelated text

Q: Who was the first human in space ?

→ **Gagarin** ✓

Unchanged

Unrelated image



→ **(unchanged)** ✓

Unchanged

Unrelated fact

Q: Which planet is known as the Red Planet?

→ **Mars** ✓

Unchanged

Irrelevant chat

Q: How are you?

→ **(unchanged)** ✓

Unchanged